LAYED PROMISE

THE JOURNEY SO FAR

2012-13: INITIAL DEVELOPMENT AND TRIALS

Feb. 2012 Anil Kakodkar Committee recommends a Radio-based Train Control Automatic Train Protection System for Indian Railways

Jan. 2013 Railway Board Committee directs the development of a full-fledged, multi-vendor interoperable train control system for less busy routes. Initial R&D tenders are awarded for 250 km, field trials start on locomotives and freight trains

2014-15: EARLY IMPLEMENTATIONS AND R&D

Core R&D is extended to auto-signalling sections, successfully demonstrating Kavach on a 15-km stretch under South Central Railway in Telangana

2016-17: PROGRESS AND POLICY FUMBLE

Feb. 2016 Field trials begin on passenger trains

May 2017 Kavach Spec. (Ver. 3.2) is approved

Jul. 2017 Railway Board mandates European Train Control System Level 1 rollout on all Golden Quadrilateral-Golden Diagonal (GQ-GD) routes, creating policy confusion

2018-20: PRODUCT APPROVAL AND EXPANSION

2018 First Kavach product (Ver. 3.2) is approved

2019 Installation for 1,200 km on South Central Railway is issued to three OEMs, adding to 265 km already under trial. Total route length now: 1,465 km

2020 Railway Board corrects policy, mandating the implementation of Train Collision Avoidance System everywhere. Work on Ver. 4 begins

2021-2024: NEW NAME AND SOME PUSH

Jul. 2021 Train Collision Avoidance System is renamed 'Kavach'

2022 Union budget allocates funds for rapid implementation across 2,000 km. Contracts are signed for 3,000 km

Feb. 2024 Trial begins in Mathura-Palwal section, optical fibre cable on 4,055 km, equipment at 273 stations and in 301 locos, installation of 356 telecom towers, trackside equipment on 1,330 route km

tion of anonymity, that the process of involving the RDSO and obtaining its approval at each stage, while keeping an eye out for the Centre of Excellence, has often meant unforeseen delays. On top of this trinity of vendors, RDSO and the Centre of Excellence, there is the Railway Board, adding another layer of bureaucratic oversight.

Soon after the Darjeeling accident, Vaishnaw reviewed the progress on the project and the ministry and, once again, stated that Kavach was to be taken up on "mission mode". Kavach has now reached its latest version (4.0) from the earlier 3.2, we are told. Those involved in the project, however, have a different take. "The development of Version 4 is delayed by almost two years, because of multiplicity of feedback and accountability. Even now, Version 4 is not yet ready," says a senior official. Even the 1,465 km where Kavach stands installed was part of an overall trial approved years ago (see *The Journey So Far*).

A LONGER WAIT

This only means that platitudes aside, the wait for a collision-free train system might be longer than expected. The plan is to complete around 6,000 km in the next two years, says Jaya. "This year, we will complete 3,000 km. Then we are planning for another 3,000 km next year. We are asking our equipment suppliers to ramp up their production, and we are helping them in that. Because this needs to be proliferated across India, gradually."

The ministry has informed Parliament that Kavach tenders have been awarded for the Delhi-Mumbai and Delhi-Howrah corridors (approximately 3,000 route km) and preparatory works, including survey, Detailed Project Report and preparation of detailed estimate on another 6,000 route km, are going on. But the way things stand today, even that seems optimistic, experts say. "Around 15,000 engines need to have Kavach. That number is now some 150 after so many years. At this rate, the total rollout might not happen in our lifetime," says L.C. Trivedi, former general manager, East Central Railway.

Even at around 1,500 km per year, it would take Indian Railways several years to cover the entire network, according to Mohammad Jamshed, former member (traffic), Railway Board, who believes railways should rope in its PSUs to implement it faster and focus not just on high-density routes but on all routes. "Kavach is not the only remedy. Concerted measures towards improving safety, already under way, need to be reassigned topmost priority," he says. "In the 21st century, no developed transportation mode can be a high-risk endeavour." Developing indigenous technological capacity is a desirable goal, but safety should be the greater priority. Choosing to catch a train in India should not be a fraught journey. ■